1 Cauchy's Mean Value Theorem

To prove L'Hôpital's Rule, we will need the following generalization of the Mean Value Theorem.

Theorem 1.1 (Cauchy's Mean Value Theorem)**.** *Let f and g be functions such that*

(i) *f and g are continuous on $[a,b]$,*

(ii) *f and g are differentiable on (a,b), and*

(iii) *$g'(x) \neq 0$ for all $x \in (a,b)$.*

Then there exists $c \in (a,b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}.$$

1. Show that in the special case in which $g(x) = x$, Theorem 1.1 is just the ordinary Mean Value Theorem.

2. Prove Theorem 1.1 by applying Rolle's Theorem to the function

$$h(x) = f(x) - f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(x)-g(a)).$$

2 L'Hôpital's Rule

Theorem 2.1 (L'Hôpital's Rule)**.** *Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that*

$$\lim_{x\to a} f(x) = 0 \quad \textit{and} \quad \lim_{x\to a} g(x) = 0$$

or that

$$\lim_{x\to a} f(x) = \pm\infty \quad \textit{and} \quad \lim_{x\to a} g(x) = \pm\infty.$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is $\pm\infty$).

Proof. We will first prove L'Hôpital's Rule for indeterminate limits of type $\frac{0}{0}$. Assume that $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ and let $L = \lim_{x\to a} \frac{f'(x)}{g'(x)}$. We need to show that $\lim_{x\to a} \frac{f(x)}{g(x)} = L$.

3. Define

$$F(x) = \begin{cases} f(x) & \text{if } x \neq a, \\ 0 & \text{if } x = a, \end{cases} \quad \text{and} \quad G(x) = \begin{cases} g(x) & \text{if } x \neq a, \\ 0 & \text{if } x = a. \end{cases}$$

   Show that F and G are both continuous on I.

4. Let $x \in I$ and $x > a$. Show that the hypotheses of Cauchy's Mean Value Theorem hold for F and G on $[a, x]$.

5. Show that there is a number y such that $a < y < x$ and

$$\frac{F'(y)}{G'(y)} = \frac{F(x)}{G(x)}.$$

6. Noting that $x \to a^+$ implies $y \to a^+$ (since $a < y < x$), use your answer above to show that

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

A similar argument shows that $\lim_{x \to a^-} \frac{f(x)}{g(x)} = L$. Therefore $\lim_{x \to a} \frac{f(x)}{g(x)} = L$, proving L'Hôpital's Rule for the case where a is finite.

7. If a is infinite, let $t = \frac{1}{x}$. Then $t \to 0^+$ as $x \to \infty$, so

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{t\to 0^+} \frac{f(1/t)}{g(1/t)}.$$

Use L'Hôpital's Rule for finite a to prove that this is equal to $\lim_{x\to\infty} \frac{f'(x)}{g'(x)}$, completing the proof for indeterminate limits of type $\frac{0}{0}$.

8. Suppose now that $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are both ∞ (or both $-\infty$).

   (a) Suppose $\lim_{x\to a} \frac{f(x)}{g(x)} = L$, where $L \neq 0, \pm\infty$. Write $L = \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{1/g(x)}{1/f(x)}$ and conclude that $L = \lim_{x\to a} \frac{f'(x)}{g'(x)}$.

(b) Now suppose $\lim_{x\to a}\frac{f(x)}{g(x)} = L = 0$. Apply the result in part (a) to the function $\frac{f(x)+g(x)}{g(x)}$ to conclude that $L = \lim_{x\to a}\frac{f'(x)}{g'(x)}$.

(c) Finally, suppose $\lim_{x\to a}\frac{f(x)}{g(x)} = L$, where $L = \infty$ (or $-\infty$). Consider $\lim_{x\to a}\frac{g(x)}{f(x)}$ to conclude that $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$. (*Hint*: Use Cauchy's Mean Value Theorem to determine the sign of $\lim_{x\to a}\frac{f'(x)}{g'(x)}$.)

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